

Verifier stack: different checks answer different failure questions

AstraBuild does not collapse provenance, software validity, geometric agreement, physical placement, and coverage into one score

History & provenance

prior object/scene/collection signatures • site transforms unchanged • SHA-256 source / protected-file chain

Artifact integrity

save → reopen round trip • finite geometry • manifest-based numeric reproduction

Geometry & placement

BVH nearest-surface residuals • ground / support / endpoint continuity • rigid instance SVD + $\det > 0$

Reviewability

camera framing / non-empty renders • qualitative overlay review • quarantine / limitation records

Completeness signal

10 cm occupancy audit • selected >1 m unexplained fraction • coverage-driven task selection

Figure 8 · Passing one layer does not imply passing another: e.g., a software-valid transform can still fail held-out geometry, and a low residual does not prove independent survey accuracy.